5.5 Arithmetic Mean

Now, let us see one of the measures of central tendency, that is the Arithmetic Mean. Consider this situation.

Mani and Ravi started collecting shells in the sea shore with an agreement to share the shells equally after collection. Finally, Mani collected 50 shells and Ravi collected 30 shells. Now, if both of them share equally, find the number of shells each one gets?

We find it using arithmetic mean or average. To find the average, add the numbers and divide by 2. Hence,

$$\text{Average} = \frac{50+30}{2} = \frac{80}{2} = 40$$

Average lies between 30 and 50.

Hence, each of them will get 40.

Thus to find the arithmetic mean (average), we have to add all the observations and divide the sum of all observations by the number of observations.

$$\text{Hence, Arithmetic Mean} = \frac{\text{Sum of all observations}}{\text{Number of observations}}.$$

Try these

Find the Arithmetic Mean or average of the following data.

(i) The study time spent by Kathir in a week is 3 hrs, 4 hrs, 5 hrs, 3 hrs, 4 hrs, 3:45 hrs; 4:15 hrs.

(ii) The marks scored by Muhil in five subjects are 75, 91, 48, 63, 51.

(iii) Money spent on vegetables for five days is ₹ 120, ₹ 80, ₹ 75, ₹ 95 and ₹ 86.

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Example 5.1 The daily wages of a worker for 10 days is as follows. Find the average income of the worker.

Day	1	2	3	4	5	6	7	8	9	10
Wages (₹)	250	350	100	400	150	270	450	320	610	750

Solution

$$\text{Arithmetic Mean} = \frac{\text{Sum of all observations}}{\text{Number of observations}}$$

$$= \frac{250+350+100+400+150+270+450+320+610+750}{10}$$

$$= \frac{3650}{10} = 365$$

Hence, the average income of the worker is ₹ 365.

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